Boolean Algebra

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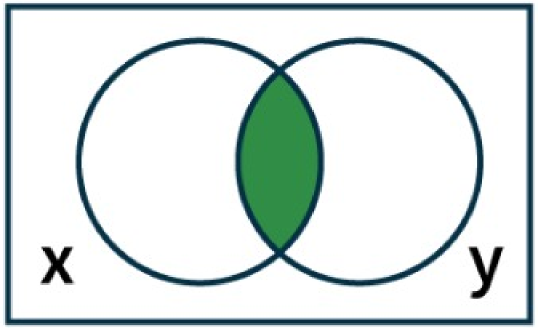
What is Boolean Algebra

Boolean algebra is a branch of mathematics and logic that deals with variables that can only take two values: **True or False** (often represented as 1 or 0).

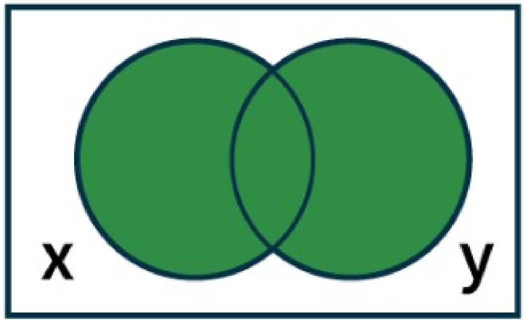
- Boolean Algebra provides a formal way to represent and manipulate logical statements and binary operations.
- It is the mathematical foundation of digital electronics, computer logic, and programming conditions.

Logical Operations

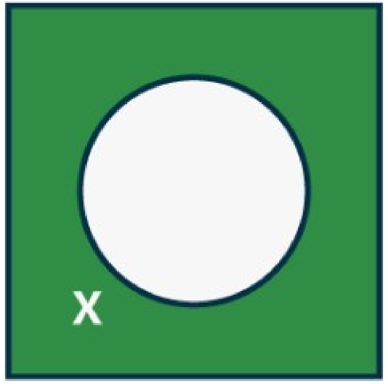
Operator	Symbol	Precedence
NOT	' (or) $\neg$	First
AND	. (or) $\wedge$	Second
OR	+ (or) $\vee$	Third



$x \wedge y$



$x \vee y$



$\neg x$

Boolean Expression and Variables

Boolean expression is an expression that produces a Boolean value when evaluated, i.e., it produces either a true value or a false value. Whereas Boolean variables are variables that store Boolean numbers.

Ex. $F=A\cdot B+C$

Boolean Identities

Identity	*Name*
$\bar{\bar{x}} = x$	Law of the double complement
$x + x = x$ $x \cdot x = x$	Idempotent laws
$x + 0 = x$ $x \cdot 1 = x$	Identity laws
$x + 1 = 1$ $x \cdot 0 = 0$	Domination laws
$x + y = y + x$ $xy = yx$	Commutative laws
$x + (y + z) = (x + y) + z$ $x(yz) = (xy)z$	Associative laws

Continue…

$x + yz = (x + y)(x + z)$ $x(y + z) = xy + xz$	Distributive laws
$\overline{(xy)} = \overline{x} + \overline{y}$ $\overline{(x + y)} = \overline{x}\,\overline{y}$	De Morgan's laws
$x + xy = x$ $x(x + y) = x$	Absorption laws
$x + \overline{x} = 1$	Unit property
$x\overline{x} = 0$	Zero property

Exercise

Ex-1: Prove the absorption law $x(x + y) = x$ using the other identities of Boolean algebra

Ex-2: Show that the distributive law $x(y + z) = xy + xz$ is valid.

Ex-3: Show that $(1 \cdot 1) + (0 \cdot 1 + 0) = 1$.

Ex-4: Use a table to express the values of each of this Boolean functions. $F(x, y, z) = xy + yz$

Duality

Every Boolean expression remains valid if we **interchange** the operations **AND (·)** and **OR (+)** and also **swap** the constants **0** and **1**.

- Replace every **+ (OR)** with **· (AND)**
- Replace every **· (AND)** with **+ (OR)**
- Replace every **0** with **1**
- Replace every **1** with **0**

Ex-1: Find the duals of $x(y + 0)$ and $x \cdot 1 + (y + z)$.
Ex-2: Find the duals of $xz + x \cdot 0 + x \cdot 1$

Sum of Products (SOP) – Disjunctive Normal Form

Sum of Product is the abbreviated form of **SOP**. Sum of product form is a form of expression in Boolean algebra in which different product terms of inputs are being summed together. This product is not arithmetical multiply but it is Boolean logical AND and the Sum is Boolean logical OR.

General Form

$$F(A, B, C, \ldots) = A'B'C + AB'C + ABC' + \ldots$$

Ex-1: Find the sum-of-products expansion for the blew function

1. $F(x, y, z) = (x + y)z$
2. $F(x, y, z) = (x + z)y$
3. $F(x, y, z) = x\,y$

K-Map(Karnaugh Map)

- A Karnaugh Map (K-Map) is a graphical tool used to simplify Boolean algebra expressions and design digital logic circuits.
- The main goal is to minimize the number of literals (variables and their complements) in a sum-of-products (SOP) or product-of-sums (POS) expression with the fewest steps and the least chance of error.
- With n variables there should be 2^n cells
- Based of Gray Code(Unit distance)

A \ B	0	1
0	0	1
1	3	2

$2^2 = 4$ cells

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

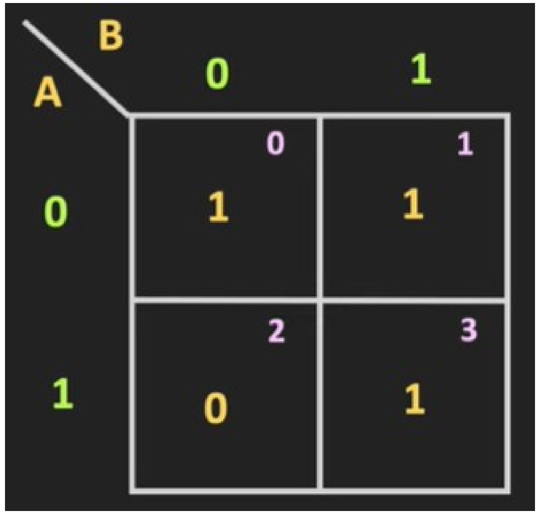
$2^3 = 8$ cells

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$2^4 = 16$ cells

Rules of Grouping

- Group only 1s.
- Groups must be powers of 2 (1, 2, 4, 8, 16, …).
- Groups must be rectangular.
- Groups can wrap around edges.
- Make the largest possible groups.
- Each 1 must be in at least one group.
- Overlapping groups are allowed.
- No diagonal grouping.
- Do not group 0s (except X if useful).
- Each group eliminates variables.
- Final SOP is OR of all groups.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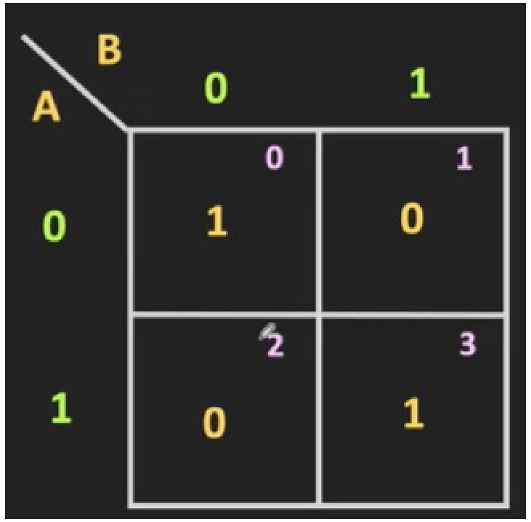
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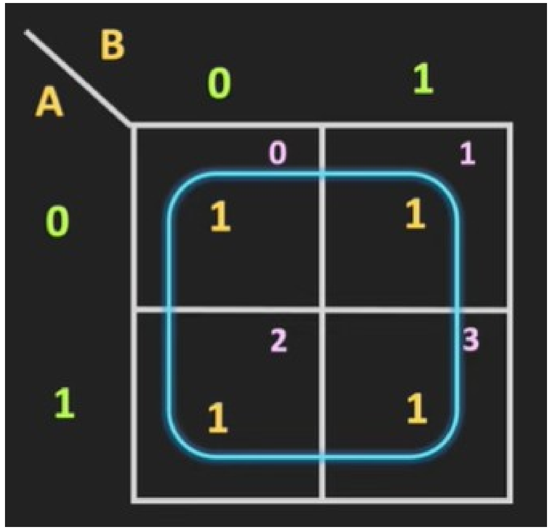
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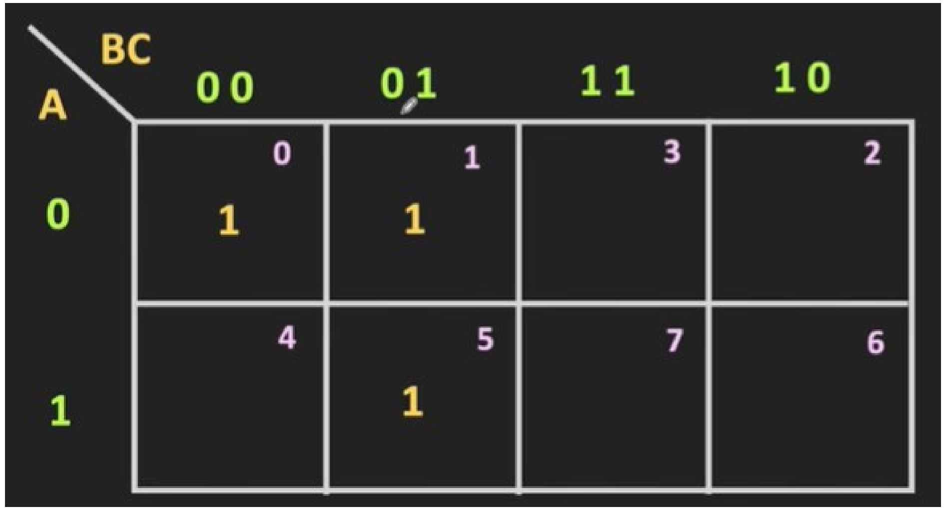
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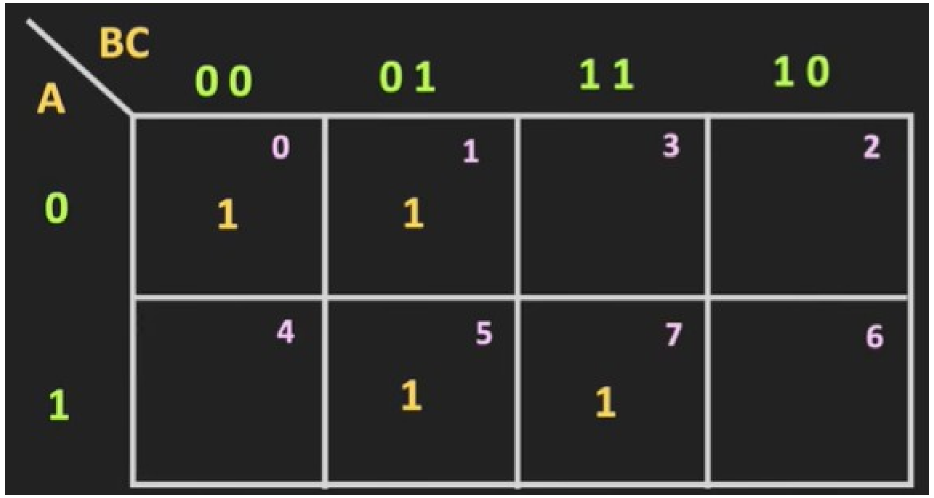
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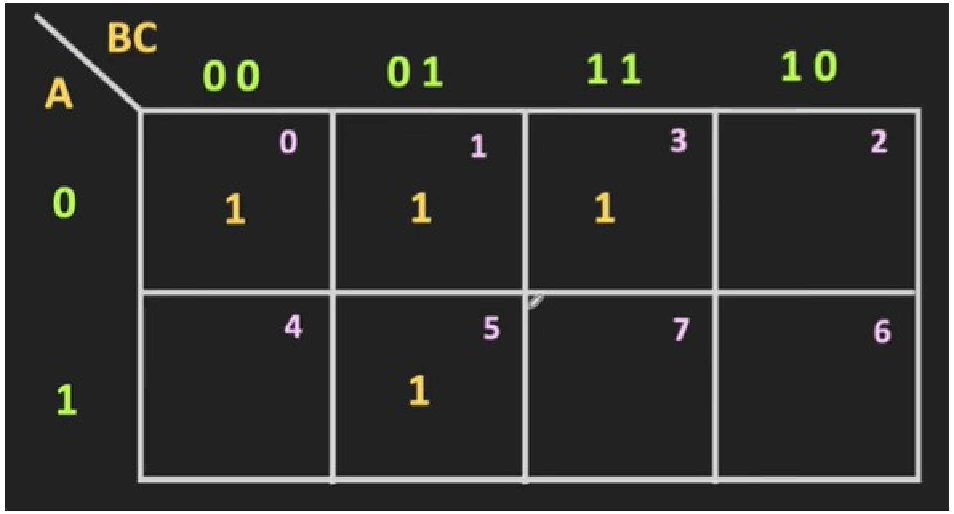
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A \ BC	00	01	11	10
0	0 1	1	3	2 1
1	4	5	7	6

A \ BC	00	01	11	10
0	0	1	3	2
1	4 1	5	7	6 1

A \ BC	00	01	11	10
0	0 1	1	3	2 1
1	4 1	5	7	6 1

A \ BC	00	01	11	10
0	0 1	1 1	3 1	2
1	4 1	5 1	7 1	6

A \ BC	00	01	11	10
0	0 1	1 1	3	2 1
1	4 1	5 1	7	6 1

Practice Problem

Use K-maps to minimize these sum-of-products expansions:

(a) $xyz + x\bar{y}\bar{z} + x\bar{y}z + \bar{x}y\bar{z}$

(b) $x\bar{y}z + x\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}yz + xy\bar{z}$

(c) $xyz + x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}y\bar{z} + xy\bar{z} + \bar{x}y\bar{z}$

(d) $xyz + x\bar{y}\bar{z} + x\bar{y}z + \bar{x}y\bar{z}$

(e) $F=\bar{A}B\bar{C}+\bar{B}C\bar{D}+\bar{A}BC\bar{D}+A\bar{B}\bar{C}$